

Section A: Aptitude & Technical MCQs

Q1. A country has a population of approximately 1,400,000,000. Which of the following is the most suitable variable declaration in C?

A) short int population = 1400000000;

B) unsigned int population = 1400000000;

C) char population = 1400000000;

D) float population = 1400000000;

Q2. What will be the output of the following C code?

```
#include <stdio.h>
int main() {
    unsigned char x = 255;
    x = x + 1;
    printf("%d", x);
    return 0;
```

A) 255

B) 256

C) 0

D) Compilation Error

Q3. What will be printed by the following program?

```
#include <stdio.h>
int main() {
    int a = 0, b = 10, c = 20;

    if (a++ && ++b) {
        c += 5;
    }

    printf("a = %d, b = %d, c = %d\n", a, b, c);
    return 0;
```

A) a = 1, b = 11, c = 25

B) a = 1, b = 10, c = 20

C) a = 0, b = 10, c = 20

D) a = 1, b = 11, c = 20

Section B: Hands-on Coding Problems

Q4. Area of a Rectangle

Define two symbolic constants:

LENGTH = 10 and BREADTH = 5.

Write a C program to calculate and print the area and perimeter of the rectangle.

Q5. Student Marks Storage

Write a C program to store the marks of a student in three subjects:

English = 75

Computer = 88

Math = 92

Calculate and display the total marks and average marks obtained.

Q6. Product Bill Calculation

A shop sells a product where: price = 499.50 quantity = 3

Write a C program to declare appropriate data types, compute the total bill amount, and print the final invoice.

Q7. Divisibility Check

Write a C program to check whether a given integer is divisible by 3 OR 5.

Input: 15

Output: Number is divisible by 3 or 5

Q8. Senior Citizen Discount Eligibility

Write a C program to check if a person qualifies for a senior citizen discount based on the following conditions:

Age $\geq$ 60

Valid ID status (id == 1)

Note: Both conditions must be true for eligibility.

Q9. Driving License Eligibility

Write a C program to determine if an applicant is eligible for a driving license. The conditions to pass are:

Age $\geq$ 18

Passed the driving test (test_passed == 1)

Has a valid ID (valid_id == 1)